

CS3340: Data Science and Management

Mapping the Sand Mafia: A Multi-Lens Data Science Analysis of Illegal Sand Mining in India

DSM Project Final Report Submission

Urvashi Balasubramaniam Manya Garg

April 26, 2026

Abstract

Illegal sand mining is one of India's most pervasive and violent forms of organized crime, yet it remains systematically under-studied and chronically under-documented. This report presents a multi-lens data science investigation that confronts, as its defining challenge, the near-complete absence of cohesive, centralized data on illegal extraction activity in India. No single authoritative database exists. Instead, evidence of the Sand Mafia is scattered across citizen field reports, journalistic archives, National Green Tribunal court filings, government district surveys, environmental sensor streams, and administrative housing records that were never designed to interoperate. Our principal accomplishment is the synthesis of seven heterogeneous data sources spanning criminal justice, housing policy, environmental monitoring, socioeconomic indicators, and real-time sensor telemetry into a unified analytical framework: the India Sand Watch platform, the National Data and Analytics Platform (NDAP), the Water Resources Information System (WRIS), the Central Pollution Control Board (CPCB), Pradhan Mantri Awas Yojana-Urban (PMAY-U) administrative records, Census 2011 socioeconomic tables, and two continuous water-quality sensors at Prayagraj. Drawing on 2,557 mining records, 91 court documents, 124 news reports, and 375 field observations spanning 2001 to 2026, we apply exploratory spatial analysis, Difference-in-Differences, Two-Stage Least Squares, synthetic control, bootstrapped mediation, NLP-based classification, and machine learning to construct, piece by piece, a picture of what illegal sand mining looks like in the absence of direct observational data. We identify two dominant mining corridors (the Gangetic plains and the Central India plateau), establish through causal inference methods that the 2015 PMAY-U launch caused a genuine increase in illegal extraction (DiD $\hat{\beta} = +1.057$, $p < 0.001$; synthetic control ex-

cess: +158% for Uttar Pradesh), and find that 100% of Bihar groundwater monitoring stations and 78% of West Bengal stations show elevated Total Dissolved Solids near mining sites. Data fragmentation is not merely a methodological inconvenience: it is a structural enabler of the Sand Mafia’s impunity, and demonstrating that a rigorous cross-sector analysis is possible despite it is itself a contribution to future research and policy design.

Contents

1	Problem Statement Formulation	4
1.1	High-Level Thematic Problem	4
1.2	Why Sand Mining Is Harmful	5
1.3	The Data Integration Challenge	6
1.4	Primary Relevant Datasets	7
1.4.1	NDAP Datasets Utilized	7
1.4.2	External Datasets Utilized	8
1.5	Exploratory Data Analysis Results	9
1.5.1	Mining Observations	9
1.5.2	Court Documents	10
1.5.3	Temporal Trends	10
2	Database System	11
2.1	Storing the Data	11
2.2	Analysis	11
2.2.1	Spatial Distribution and Clustering (spatial_analysis.ipynb)	11
2.2.2	Construction Demand and Causal Inference (construction_mining_analysis.ipynb)	16
2.2.3	Socioeconomic Correlates (police_literacy_marginal_workers_agriculture.ipynb and household_and_crime.ipynb)	20
2.2.4	NLP-Based Classification of Illegality (classification.ipynb)	23
2.2.5	Water Quality: Ganga and Sangam Sensor Analysis (spatial_analysis.ipynb, Section 11)	25
2.3	Current State of the Issue	27
3	Key Quantitative Findings and Significant Associations	29
3.1	Causal Evidence: Construction Demand Drives Illegal Mining	29
3.2	Spatial Finding: Groundwater Contamination Near Mining Sites	30
3.3	Socioeconomic Associations	31
3.4	Environmental Time-Series Findings	31
3.5	Classification and Enforcement Gap	32

4	Recommendations and Policy Suggestions	32
4.1	For the Central Government	32
4.2	For State Governments	33
4.3	For Judicial and Regulatory Bodies	33
4.4	For Civil Society and Research	34

1 Problem Statement Formulation

1.1 High-Level Thematic Problem

Sand is the second-most exploited natural resource in the world after water. It is the primary input for concrete, glass, asphalt, and electronics manufacturing, making it indispensable to modern construction and industry. The United Nations Environment Programme (UNEP) estimates that more than 50 billion tonnes of sand and gravel are extracted globally every year, a figure that dwarfs the replenishment capacity of rivers and coastal systems [1]. Demand is only expected to intensify: the United Nations estimates that by 2030, the world will have 40 megacities (cities with populations exceeding 10 million), compared to 31 today [2], each requiring enormous quantities of construction-grade aggregates. India sits at the center of this pressure: it is one of the fastest-urbanizing nations on earth, with a government-mandated housing program (the Pradhan Mantri Awas Yojana-Urban, or PMAY-U) that has sanctioned nearly 12 million new houses since its launch in 2015.

Because legally permitted sand extraction struggles to meet this accelerating demand, a shadow supply chain has emerged. India’s “Sand Mafia” is an informal network of organized criminal groups that extract, transport, and sell sand illegally from rivers, floodplains, and coastal areas. Their operations are frequently accompanied by violence: between 2019 and 2023, documented incidents include the deaths of government officials, journalists, and civilians who attempted to report or obstruct mining activities [3]. Despite the scale and severity of this phenomenon, academic and policy research on India’s Sand Mafia remains sparse. Most existing literature treats it as a localized governance failure rather than a systemic, nationally distributed organized crime problem with measurable structural drivers.

This report attempts to partially fill that gap. We treat illegal sand mining not as an isolated criminal event but as a complex socioeconomic outcome shaped by construction demand, poverty, enforcement capacity, environmental conditions, and geographic opportunity. Our analysis proceeds through four thematic lenses:

1. **Geographic lens:** Where is illegal sand mining concentrated, and does it form spatial clusters that suggest coordinated criminal networks rather than opportunistic individual actors?
2. **Demand lens:** Can we demonstrate a causal link between large-scale government construction programs and the intensification of illegal extraction?
3. **Socioeconomic lens:** Are states with higher rates of poverty, marginal employment, and poor housing conditions more prone to sustained illegal mining activity?
4. **Environmental lens:** Does proximity to active sand mining sites correlate with measurable degradation in groundwater quality and air particulate levels?

Our end goal is not merely descriptive. We aim to identify which policy levers (enforcement, legal reform, environmental monitoring, and alternative livelihoods) are most likely to disrupt illegal mining operations, and to present this analysis in a form that can directly inform government action.

1.2 Why Sand Mining Is Harmful

Before proceeding to the data, it is worth grounding the problem in its environmental and human consequences, because the urgency of these consequences is what justifies the analytical effort.

Ecological destruction. Rivers are not static containers of water. They are dynamic systems in which sand and gravel serve as habitat for thousands of species, regulate streamflow, anchor riverbanks, and filter groundwater. When sand is removed from riverbeds faster than natural deposition can replace it (a process called *riverbed incision*), channels deepen and banks become unstable. This triggers a cascade: bridges lose their foundations, floodplains no longer receive seasonal sediment, and the water table drops because the natural sponge layer beneath the river is removed. Studies of heavily mined rivers in India have documented bank collapses, loss of aquatic biodiversity, and the drying up of small tributaries [4].

Groundwater contamination. Sand mining disturbs the mineral-rich sediment layer of riverbeds, releasing dissolved solids (including heavy metals) into the water column. Total Dissolved Solids (TDS), the measure of the combined content of all inorganic and organic substances dissolved in water, are a standard proxy for this contamination. Our analysis of Water Resources Information System (WRIS) groundwater quality data across India, examined in Section 2.2.1, finds that stations closer to active mining sites tend to show elevated TDS readings, with the effect most pronounced in Bihar and West Bengal.

Air quality degradation. The transport of sand by truck, tractor, and barge generates large quantities of coarse particulate matter. PM10 and PM2.5, the two standard regulatory metrics for airborne particles (referring to particles with diameters of 10 and 2.5 micrometers or less, respectively), have been linked to respiratory disease, cardiovascular damage, and premature death in exposed populations. Our spatial regression analysis finds that PM10 is the most strongly correlated environmental variable with mining intensity at the state level.

Violence and impunity. The Sand Mafia operates in a context of systemic legal impunity. Court documents in our dataset reveal patterns in which local officials are allegedly bribed, whistleblowers are threatened, and environmental regulations are routinely violated without sanction. The gap between observed mining intensity and the rate of court-recognized legal cases (analyzed in Section 2.2.4) is itself a measure of this impunity.

Social inequality. Illegal sand mining is not uniformly distributed across society. It cor-

relates highly with the geographic concentration of marginal workers in river-adjacent and resource-dependent areas, where livelihoods are most vulnerable to environmental disruption. The communities whose rivers and groundwater are depleted receive no compensation. This makes illegal sand mining both an environmental problem and a question of social justice.

1.3 The Data Integration Challenge

The central methodological challenge of this project is one that no amount of analytical sophistication can fully overcome: *there is no single, authoritative, comprehensive dataset on illegal sand mining in India*. Unlike conventional crime statistics, which are collected by a centralized agency (the NCRB) and published in a standardized annual format, sand mining incidents are scattered across field observation networks, journalistic archives, court filings, government district surveys, and environmental monitoring databases that were never designed to interoperate.

This fragmentation is not accidental. The Sand Mafia thrives in informational darkness: the absence of systematic documentation is itself a governance failure that perpetuates impunity. When there is no official count of mining incidents, there is no official accountability for failing to address them.

Our response to this challenge is the defining accomplishment of this project: the assembly of a unified analytical framework from seven fundamentally different data sources.

1. **India Sand Watch (Veditum India Foundation):** 2,557 incident records from field observers, journalists, court documents, and government district surveys, spanning 2001 to 2026. This is the only publicly available national-scale mining incident database in India. Its coverage is uneven (Bihar and Madhya Pradesh are overrepresented; northeastern states are nearly absent), and its records are heterogeneous in quality and completeness.
2. **National Data and Analytics Platform (NDAP):** Official government statistical series from the NCRB (crime and police personnel), Census 2011 (population, literacy, marginal workers, household conditions), CPCB (air quality), and PMAY-U (housing construction progress and central assistance allocations).
3. **Water Resources Information System (WRIS):** 7,190 groundwater monitoring station records with location coordinates and Total Dissolved Solids (TDS) measurements, spatially linked to the nearest confirmed mining point for Geographically Weighted Regression.
4. **Central Pollution Control Board (CPCB):** State-level annual average PM2.5 and PM10 concentrations from the national ambient air quality monitoring network.

5. **PMAY-U Administrative Records:** State-level housing sanctions, groundings, completions, and central government assistance allocations, used as the primary treatment variable in the causal analyses.
6. **Census 2011 Socioeconomic Tables:** State-level population, literacy rates, marginal worker proportions, and household condition data (good, liveable, dilapidated), used as socioeconomic covariates.
7. **Prayagraj Water Quality Sensors:** Two continuous monitoring buoys capturing approximately 99,000 one-minute readings of dissolved oxygen, pH, oxidation-reduction potential, conductivity, temperature, and a composite Water Quality Index (WQI) between January 2019 and February 2020.

Each of these datasets uses different geographic granularities (some state-level, some district-level, some point-level), different temporal resolutions (annual, monthly, daily, per-minute), different state name conventions, and different units and measurement standards. A substantial portion of our technical effort was devoted to making these datasets speak to each other at all, including a systematic Unicode normalization pipeline, a manually curated state name correction dictionary, and spatial join procedures linking point-level sensor and mining data to administrative polygon boundaries.

The result is not a perfect dataset. It is the most complete picture of illegal sand mining in India that can currently be assembled from publicly available sources, and demonstrating that it is possible to derive meaningful systemic analysis from these fragments is a contribution to future research in this area.

1.4 Primary Relevant Datasets

Our dataset is assembled from two primary sources: the India Sand Watch platform maintained by Veditum India Foundation, and the National Data and Analytics Platform (NDAP) operated by the Government of India.

1.4.1 NDAP Datasets Utilized

The National Data and Analytics Platform aggregates official government statistical series from multiple ministries and agencies. The following NDAP datasets were used in this analysis:

Crime in India (NCRB). The National Crime Records Bureau (NCRB) publishes annual state-level crime statistics under the Indian Penal Code (IPC). We use the “cognizable crimes reported” rate per 100,000 population as a proxy for the overall law enforcement environment in each state. A cognizable crime is one for which a police officer may arrest

without a warrant under Indian law. The dataset covers 2022 data across all 36 states and Union Territories.

Police Personnel (NCRB). State-level data on the total number of authorized and actual police personnel, disaggregated by rank (constable, sub-inspector, inspector, etc.). We aggregate these into a single “total police strength” variable and compute a mining-incidents-per-police-personnel ratio to assess enforcement capacity relative to mining activity.

Household Conditions (Census). State-level data on the proportion of households classified as being in good, liveable, or dilapidated condition. This serves as a proxy for material poverty and unmet housing needs.

Population and Literacy (Census 2011). State-level data on total population, literate population (used to compute literacy rate), and the proportion of “marginal workers” (a Census category referring to individuals who work fewer than 183 days per year, indicating unstable, seasonal, or informal employment).

PM2.5 and PM10 Air Quality (CPCB). Annual average concentrations of fine and coarse particulate matter measured at state-level monitoring stations, compiled from the Central Pollution Control Board (CPCB) database.

Groundwater Quality (WRIS). The Water Resources Information System maintains a national network of groundwater monitoring stations. We use station-level latitude, longitude, and Total Dissolved Solids (TDS) measurements, aggregated by state and by individual station for the Geographically Weighted Regression analysis described in Section 2.2.1.

PMAY-U Construction Data. State-level and district-level data on houses sanctioned, grounded, and completed under the Pradhan Mantri Awas Yojana-Urban scheme, as well as the central government assistance allocation (in crore rupees) to each state. This is the primary measure of construction demand in our causal analysis.

1.4.2 External Datasets Utilized

India Sand Watch (Veditum). This is our primary source for mining incidents. It contains four sub-datasets:

- *Mining observations* (375 records): Field reports submitted by citizen observers, journalists, and researchers, containing the location, date, type of machinery used, and descriptive notes for each observed mining incident.
- *Court documents* (91 records): Legal filings and judgments from the National Green Tribunal (NGT), High Courts, and the Supreme Court of India related to sand mining cases, with metadata including the state, document type, and a text summary.

- *News reports* (124 records): Media articles about sand mining incidents, organized by state.
- *District survey reports* (91 records): Official government district surveys mandated under the Minor Minerals Rules, containing structured assessments of sand availability and extraction in each district.
- *Unified geo data* (2,212 verified records): A consolidated dataset of all mining incidents with latitude and longitude coordinates, combining the above sources with additional records.

Ganga and Sangam water quality sensors. Two continuous water quality monitoring buoys deployed at Prayagraj, Uttar Pradesh, capturing approximately 46,528 and 52,363 one-minute readings respectively between January 2019 and February 2020. Variables include Dissolved Oxygen (DO), pH, Oxidation-Reduction Potential (ORP), conductivity (in microsiemens per centimeter), temperature, and a composite Water Quality Index (WQI).

India administrative GeoJSON files. State and district-level polygon shapefiles from the GADM global administrative areas database, used as the geographic backbone for all choropleth maps, spatial weight matrices, and spatial regression models.

1.5 Exploratory Data Analysis Results

1.5.1 Mining Observations

The 375 field observation records span 2016 to 2023, with the majority concentrated after 2017. Bihar is the single most-reported state with 193 observations, followed by Uttar Pradesh (48), West Bengal (48), and Madhya Pradesh (43). This distribution is notably skewed: Bihar alone accounts for 51% of all field observations. However, the broader unified geo dataset (2,212 records) tells a different story at the state level: Madhya Pradesh leads with 369 incidents, followed by Maharashtra (210), Uttar Pradesh (130), and Tamil Nadu (125). The divergence between observation data and aggregated incident data reflects differences in data collection methodology: field observers are concentrated in certain regions, while the aggregated data incorporates news reports, court documents, and survey data that are more uniformly distributed.

A key early finding is the severe mismatch between states in terms of “formal” versus “informal” reporting. Bihar has 193 informal field observations but very few corresponding court cases. Madhya Pradesh, by contrast, has 279 news-level incidents and a substantial body of NGT court filings. This asymmetry hints at differential legal enforcement, and motivates the classification analysis in Section 2.2.4.



Figure 1: Field observation counts by state (India Sand Watch). Bihar accounts for 193 of 375 total observations (51%), reflecting the geographic concentration of the citizen observer network rather than necessarily the true distribution of mining activity.

1.5.2 Court Documents

The 91 court documents span cases from the Supreme Court, various High Courts, and the National Green Tribunal. Topic modeling (using Latent Dirichlet Allocation, a technique that identifies recurring thematic groups of words in a collection of documents) on the Madhya Pradesh subset identified five dominant topic clusters: (1) illegal mining and environmental clearance violations, (2) Sand Mining Rules and Minerals Act provisions, (3) NGT orders and committee reports, (4) district-level enforcement actions, and (5) applications for lease renewals. Pattern matching using regular expressions (pattern-matching rules applied to text) across all 91 documents found that 63 contained explicit references to “illegal mining,” 26 referenced a failure to obtain environmental clearance, and 22 cited environmental violations; collectively these patterns suggest that the legal system is primarily engaged with environmental rather than criminal dimensions of the problem.

1.5.3 Temporal Trends

Mining incidents in the unified dataset show a dramatic and statistically significant increase beginning in 2015, the year PMAY-U was launched. Pre-2015 incidents average 13 per year; post-2015 incidents average 191 per year: a $14.7\times$ increase. This temporal break is the starting point for the causal analysis in Section 2.2.2.

2 Database System

2.1 Storing the Data

Given the diverse and multi-format nature of our data sources (CSV files from NDAP, GeoJSON shapefiles, CSV exports from India Sand Watch, and raw sensor logs), we adopted a hybrid storage architecture appropriate for a research-scale project.

The primary relational store is a PostgreSQL database hosted on Supabase, a cloud-hosted open-source database platform. All mining observation, court document, news report, and district survey data are stored in a unified table (`no_mobs_unified_geo_data`) with standardized column schemas. Geographic data (state and district polygon shapefiles) are managed as GeoJSON files read directly into Python via the GeoPandas library (an extension of the Pandas data analysis library that can handle geographic geometries). Environmental datasets (air quality, groundwater quality, sensor data) are stored as flat CSV files and joined on normalized state names at query time.

A key challenge throughout this project was state name normalization. Official government datasets, India Sand Watch records, and geographic shapefiles use inconsistent spellings and abbreviations for the same administrative unit (for example, “Andaman & Nicobar,” “Andaman and Nicobar Islands,” and “Andaman Nicobar”). We implemented a systematic normalization pipeline using Unicode normalization (to remove diacritics), lowercase conversion, punctuation stripping, and a manually curated correction dictionary mapping known variant spellings to a canonical form. This pipeline was applied uniformly across all datasets before any merge operation.

For the spatial dashboard handoff (documented in `dashboard_handoff_resource.ipynb`), all analysis outputs were serialized as GeoJSON layer files and a JSON manifest, providing us with a well-defined interface independent of the analysis codebase.

2.2 Analysis

Our analysis is organized into five interconnected modules, each corresponding to a specific notebook and a specific analytical question.

2.2.1 Spatial Distribution and Clustering (`spatial_analysis.ipynb`)

Kernel Density Estimation. We first constructed a continuous spatial density surface over India using Kernel Density Estimation (KDE). KDE is a non-parametric statistical technique that estimates the probability density of a variable (in this case, mining incidents) at each point in a geographic area, by placing a smooth “kernel” (a mathematically defined bump) at each observed data point and summing the overlapping kernels. Unlike a simple

dot map, KDE produces a continuous surface that reveals where events are most densely concentrated.

Using 373 geo-coded mining points from the unified dataset (the subset with valid latitude and longitude coordinates), KDE with a bandwidth of 0.08 degrees identifies two distinct concentration corridors:

1. **The Gangetic Plains corridor:** A high-density zone stretching along the Ganga river and its tributaries through Bihar, eastern Uttar Pradesh, and West Bengal. In Bihar alone, the Ganga river accounts for 142 of 268 total records, with the Patna district contributing 105 records: the single hottest geographic concentration in the country.
2. **The Central India Plateau corridor:** A high-density zone across Madhya Pradesh and Rajasthan, driven primarily by mining on the Chambal and Narmada rivers. Morena district in Madhya Pradesh has the highest district-level count nationally at 81 records, reflecting the Chambal’s relatively remote and politically sensitive geography.

We also constructed a “mining-to-monitoring ratio” map comparing mining incident density against groundwater monitoring station density. This map reveals that northeastern districts of India have high mining incident density but relatively low monitoring station coverage, identifying probable under-detection zones where the true scale of illegal extraction likely exceeds what the data capture.

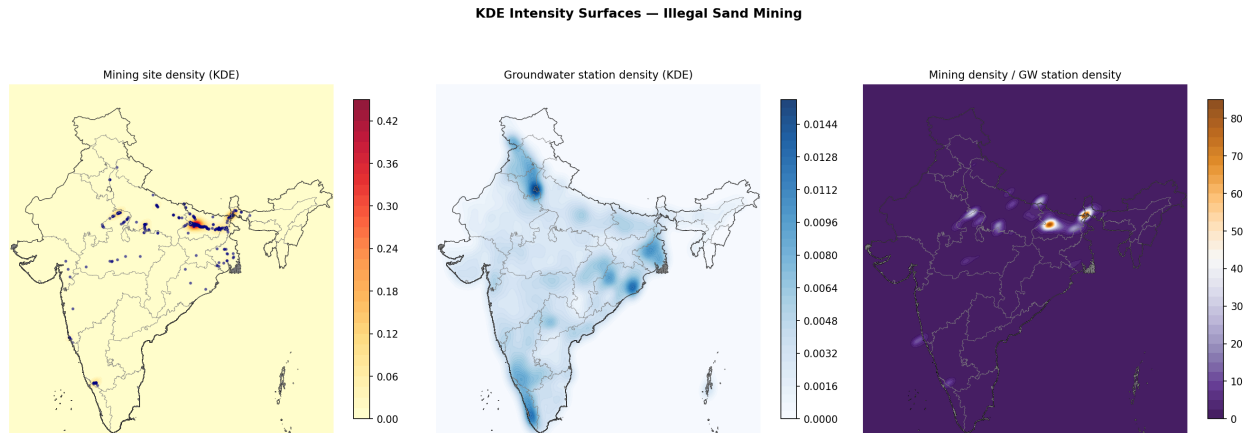


Figure 2: KDE intensity surfaces over India. Left: confirmed mining site density (bandwidth = 0.08°), showing two concentration corridors along the Gangetic plains and the Central India plateau. Center: groundwater monitoring station density. Right: mining-to-monitoring ratio, identifying northeastern districts as probable under-detection zones where illegal activity outpaces observational coverage.

Global Moran’s I. To test whether the spatial distribution of mining is random or exhibits geographic clustering, we computed Moran’s I, a measure of spatial autocorrelation rang-

ing from -1 (perfect dispersion) through 0 (complete spatial randomness) to $+1$ (perfect clustering). We defined spatial neighbors using a Queen contiguity weights matrix, meaning that two states are considered neighbors if they share at least a single boundary point. The weights matrix is row-standardized so that the weights of each state’s neighbors sum to one.

The global Moran’s I for state-level mining counts is $I = 0.121$ ($z = 1.29$, $p = 0.108$ under 999 permutations). This represents a marginal positive clustering tendency (high-mining states tend to be adjacent to other high-mining states), but the result falls just below the conventional statistical significance threshold of $\alpha = 0.05$. This means we cannot conclusively reject the null hypothesis of spatial randomness at the state level, though the positive direction of the statistic is consistent with coordinated cross-border criminal networks.

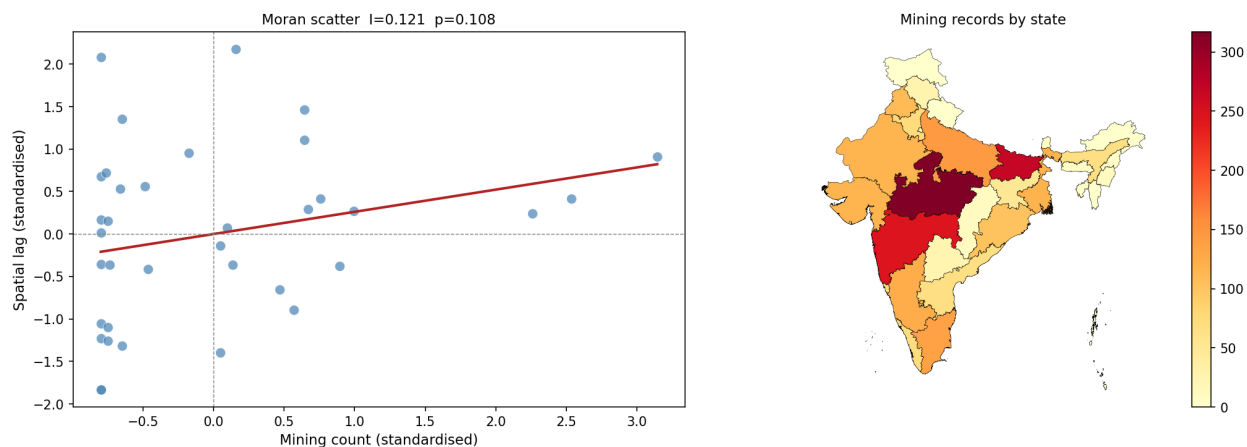


Figure 3: Global Moran’s I analysis. Left: Moran scatter plot ($I = 0.121$, $p = 0.108$); the positive regression slope indicates mild spatial clustering. Right: state-level choropleth of total mining records showing the Gangetic corridor and Central India plateau as the primary concentration zones.

LISA Hotspot Analysis. Local Indicators of Spatial Association (LISA) decompose the global Moran’s I into state-level contributions, allowing us to identify specific geographic hotspots and coldspots. Under this method, states are classified into four quadrant types: High-High (HH, a high-mining state surrounded by other high-mining states), Low-Low (LL), High-Low (HL, an isolated high-mining state surrounded by low-mining neighbors), and Low-High (LH).

At $p < 0.05$ (under 999 permutations), the following classifications emerge:

- **HH hotspot states:** Goa, Gujarat, Madhya Pradesh, and Rajasthan: a geographically contiguous belt indicating a cross-border enforcement problem. These states share rivers (most importantly the Chambal) and likely share criminal networks.
- **HL spatial outlier:** Assam, a high-mining state isolated from the main concentration

belt, suggesting locally specific drivers (likely the Brahmaputra river and its tributaries) rather than network spillovers from neighboring states.

- **LH spatial outliers:** Chhattisgarh, Dadra and Nagar Haveli, and Jharkhand: low-mining states surrounded by relatively higher-mining neighbors.
- **LL coldspots:** Andaman and Nicobar Islands and Lakshadweep, remote island territories with negligible mining activity.

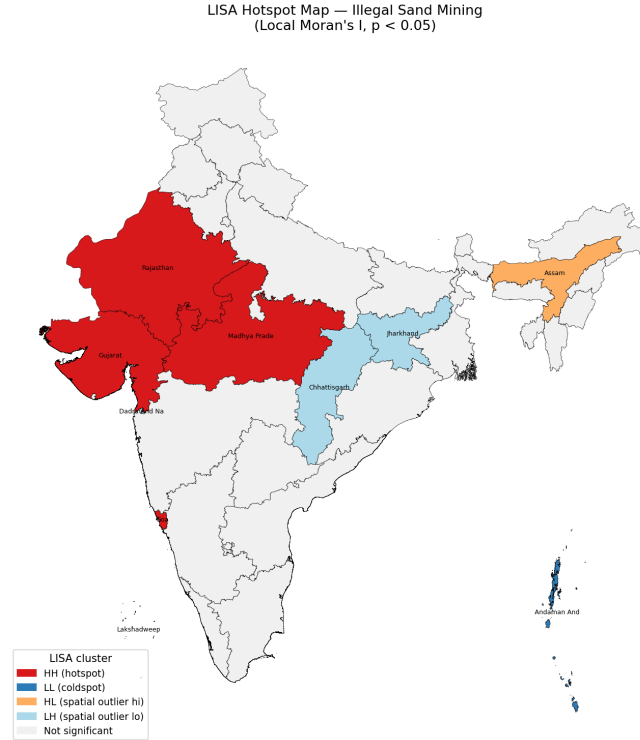


Figure 4: LISA hotspot map ($p < 0.05$, 999 permutations). The red High-High cluster (Goa, Gujarat, Madhya Pradesh, Rajasthan) forms a geographically contiguous belt sharing the Chambal and Narmada river systems. The orange High-Low outlier (Assam) reflects locally specific Brahmaputra basin drivers. Teal Low-High outliers (Chhattisgarh, Jharkhand) sit adjacent to higher-mining neighbours.

Spatial Regression Models. We estimated three regression models to assess whether crime rate, PM2.5 levels, and PM10 levels predict state-level mining intensity:

An *Ordinary Least Squares* (OLS) model serves as the baseline: it estimates the linear relationship between mining intensity (log-transformed) and the predictors without accounting for the geographic relationships between states. The OLS model achieves $R^2 = 0.158$, with PM10 showing the largest coefficient ($\beta = +1.17$, $p = 0.123$), though no predictor reaches individual statistical significance. The low R^2 indicates that state-level environmental and crime variables explain only a modest fraction of the variance in mining intensity. This is not

surprising: geography (which rivers a state contains) and political economy (which networks have entrenched themselves) are likely more important determinants.

To test whether spatial dependence among states biases the OLS estimates, we computed Lagrange Multiplier tests for spatial lag dependence and spatial error dependence. Neither test reaches significance ($p = 0.98$ for the lag test, $p = 0.74$ for the error test), so OLS is not strongly violated. We nevertheless estimate a *Spatial Lag Model* (SLM), which adds a term for the weighted average of neighboring states' mining intensity as a predictor, and a *Spatial Error Model* (SEM), which assumes spatial correlation enters through the error term (unexplained variation) rather than the mean. Neither model improves substantially on OLS (AIC of 97.97 and 95.81 respectively, versus 95.97 for OLS), consistent with the non-significant Moran's I.

Geographically Weighted Regression (GWR). Standard regression assumes that the relationship between predictors and outcomes is the same everywhere. Geographically Weighted Regression relaxes this assumption by allowing regression coefficients to vary across space: at each location, a separate regression is estimated using a locally weighted subset of nearby observations, where closer observations receive higher weights. This produces a map of locally varying coefficients rather than a single global estimate.

We applied GWR to the relationship between groundwater quality (log-transformed Total Dissolved Solids, TDS) at 7,190 WRIS monitoring stations and each station's distance to the nearest confirmed mining point. A negative local GWR coefficient on this distance variable means that closer proximity to mining is associated with higher TDS: the expected contamination signal. Globally, the GWR achieves $R^2 = 0.545$, a substantial improvement over OLS, indicating that the relationship between mining proximity and water quality is highly spatially heterogeneous.

The state-level breakdown of local GWR coefficients reveals striking regional variation. In Bihar, 100% of monitoring stations show negative coefficients (closer to mining = worse water quality), with a median $\beta = -0.346$. In West Bengal, 78% of stations show negative coefficients (median $\beta = -0.505$). In Madhya Pradesh, only 39% show negative coefficients (median $\beta = +0.215$), suggesting that the naturally mineral-rich hard-rock geology of the Chambal basin masks the contamination signal from mining.

Random Forest Classification. To identify which spatial and environmental features are most predictive of whether a location is a mining site (as opposed to a random background location within India), we trained a Random Forest classifier (an ensemble of many individual decision trees that vote on the classification of each input). Using 373 confirmed mining points and 357 randomly generated background points within India's boundaries as training data, the model achieves a 5-fold cross-validated ROC-AUC (Receiver Operating Characteristic Area Under the Curve, a standard measure of classifier performance ranging from 0.5

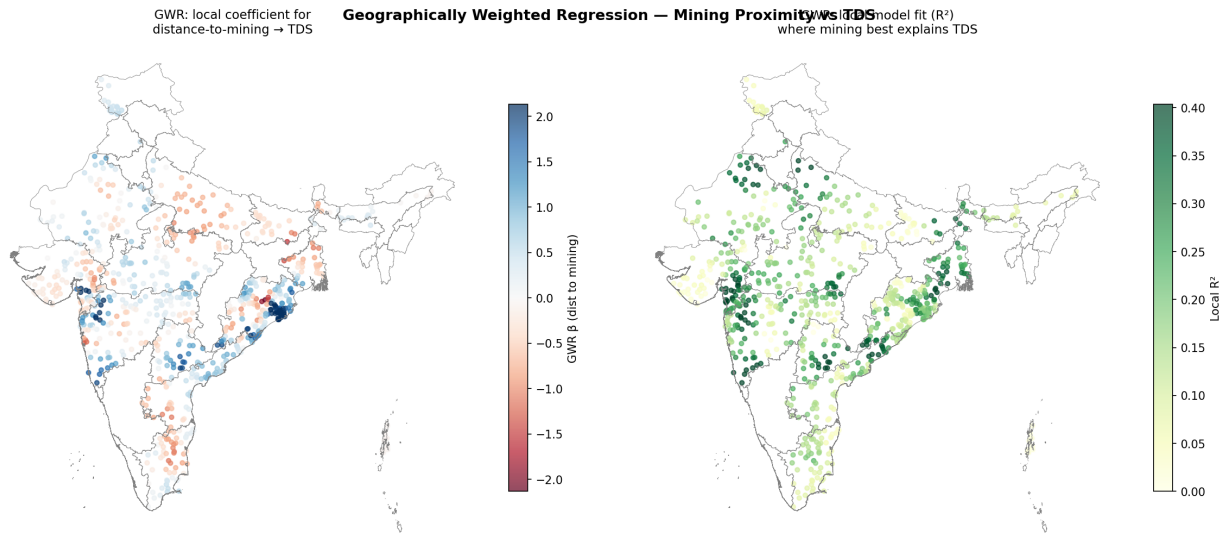


Figure 5: GWR local coefficients for distance-to-nearest-mining-site predicting $\log(\text{TDS})$ at 7,190 WRIS groundwater stations. Left: negative (blue/dark) values indicate that proximity to mining correlates with elevated dissolved solids; the signal is concentrated in the Gangetic corridor (Bihar, West Bengal). Right: local R^2 surface, confirming that the mining-water quality relationship is strongest along the Ganga and its tributaries.

for random chance to 1.0 for perfect classification) of 0.965 ± 0.014 . The feature importance ranking (measured by permutation importance, which assesses how much model performance degrades when each feature is randomly shuffled) places longitude and latitude first and second (confirming the strong geographic patterning evident in the KDE maps), followed by crime rate (third), PM10 (fourth), and PM2.5 (fifth). The fact that crime rate ranks ahead of both air quality variables is notable: it suggests that the enforcement environment is at least as strong a co-determinant of mining location as environmental conditions.

2.2.2 Construction Demand and Causal Inference (construction_mining_analysis.ipynb)

This section asks the most policy-relevant question in our analysis: did the 2015 launch of the PMAY-U housing scheme cause an increase in illegal sand mining, or is the observed temporal correlation driven by confounding factors (for example, improved data collection infrastructure after 2015)?

Temporal Analysis. The mining time series shows a clear structural break around 2015. Pre-2015 annual averages are 13 incidents per year; post-2015 averages are 191 incidents per year. A Mann-Whitney U test (a non-parametric test that compares the medians of two distributions without assuming they are normally distributed) confirms this difference is statistically significant ($p = 0.0011$). However, a Chow test for a structural break (a test that asks whether the regression slope and intercept change significantly at a specified breakpoint)

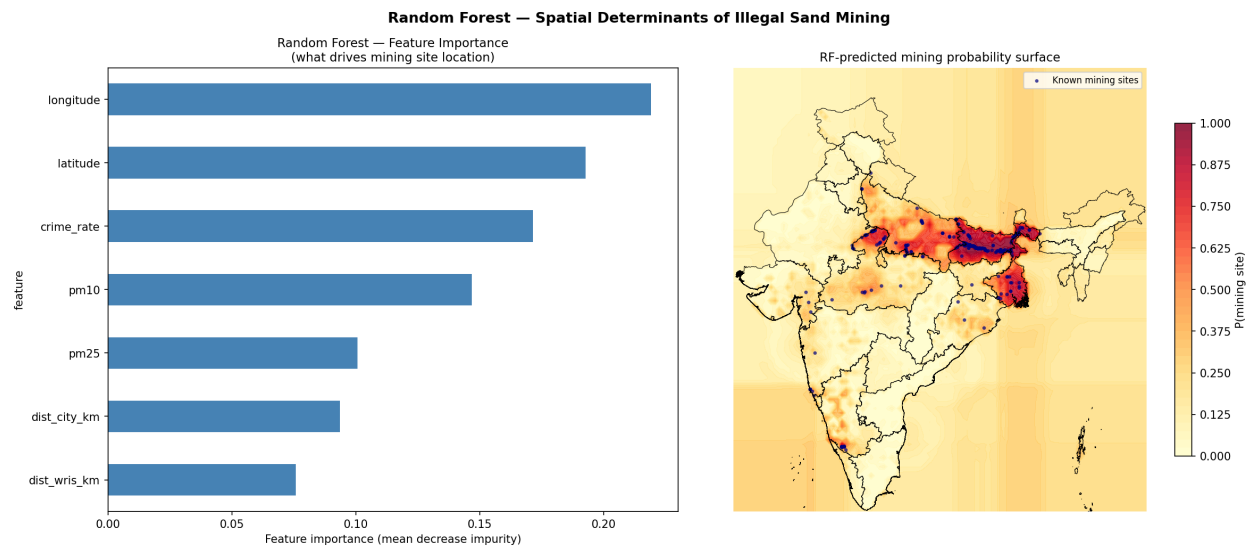


Figure 6: Random Forest spatial determinants of illegal sand mining. Left: feature importance (mean decrease in impurity). Longitude and latitude rank first and second, confirming strong geographic patterning; crime rate ranks third, ahead of PM10 and PM2.5. Right: RF-predicted mining probability surface, closely matching the KDE density map and highlighting the Gangetic and Central India corridors.

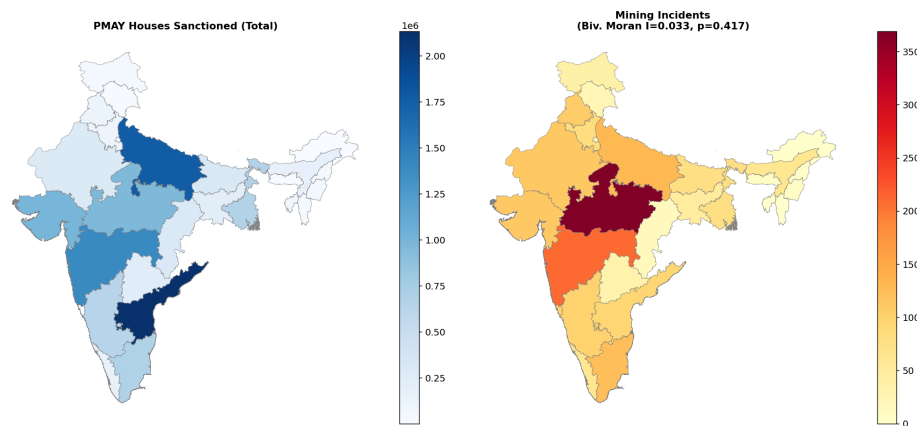


Figure 7: Side-by-side state choropleths of PMAY houses sanctioned (left) and total mining incidents (right). States with large PMAY allocations overlap substantially with high-mining states. Bivariate Moran's $I = +0.033$ ($p = 0.417$), indicating spatial co-clustering that falls just short of conventional significance.

does not reach significance at the 2015 breakpoint ($F = 0.44$, $p = 0.655$), suggesting that while levels increased, the rate of change was already trending upward before 2015.

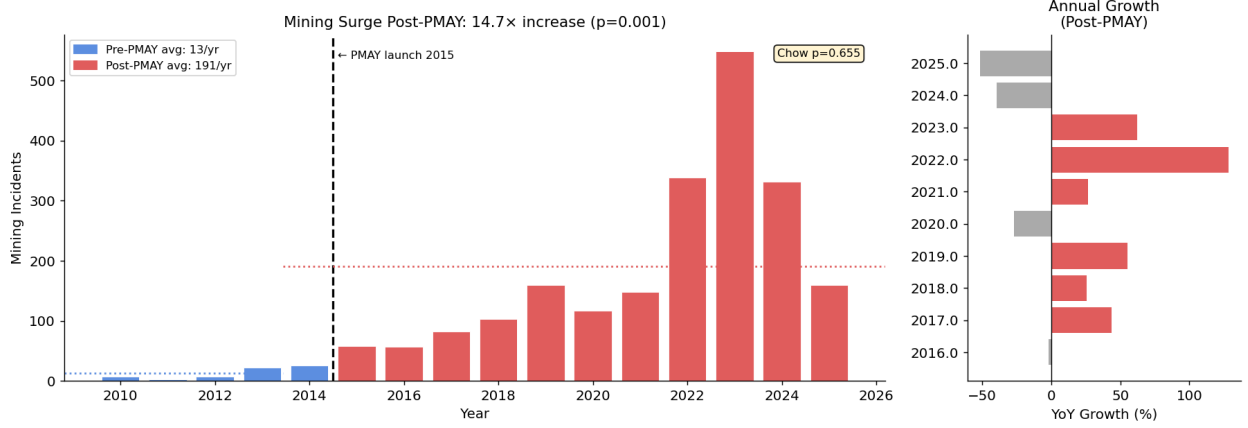


Figure 8: Annual mining incident counts (left) and year-on-year growth rates post-2015 (right). Pre-PMAY average: 13 incidents/year (blue). Post-PMAY average: 191 incidents/year (red), a $14.7\times$ surge. The Chow test ($p = 0.655$) does not find a discrete structural break, suggesting a gradual acceleration rather than an abrupt step change at the policy launch.

Difference-in-Differences. To move closer to a causal claim, we implement a Difference-in-Differences (DiD) design. This method compares the change in outcomes before and after a policy intervention in a “treated” group (states with high PMAY-U allocations) against the change in a “control” group (states with low allocations), under the assumption that both groups would have followed parallel trends in the absence of the program. Treatment is defined as being above the median in PMAY houses sanctioned per state.

We estimate a Two-Way Fixed Effects (TWFE) DiD model, which simultaneously controls for time-invariant differences between states (state fixed effects) and common time trends across all states (year fixed effects), thereby isolating the treatment effect:

$$\log(\text{mining}_{it}) = \beta \cdot (\text{Treated}_i \times \text{Post}_t) + \alpha_i + \gamma_t + \varepsilon_{it} \quad (1)$$

where α_i represents state fixed effects and γ_t represents year fixed effects. The estimated DiD coefficient is $\hat{\beta} = +1.057$ ($p < 0.001$, 95% CI: [0.855, 1.259]), implying that high-PMAY states experienced 187.7% more mining post-2015 than low-PMAY states, relative to their pre-2015 difference. An event study design (which estimates separate treatment effects for each year relative to the PMAY launch) shows no significant pre-2015 differential trends between treated and control states ($F = 0.406$, $p = 0.805$), supporting the parallel trends assumption.

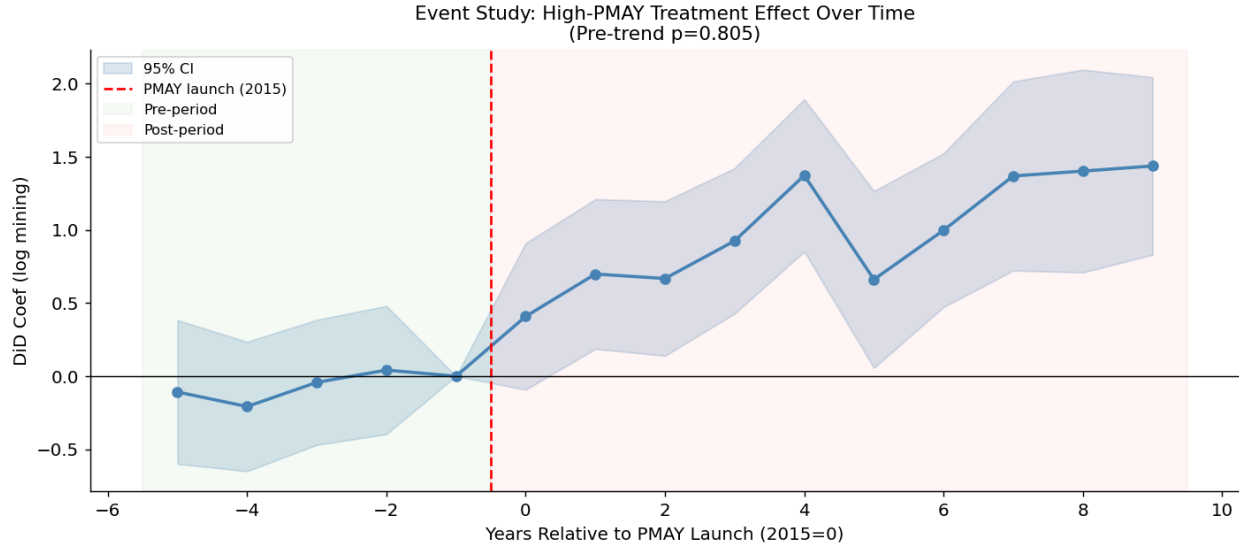


Figure 9: Event study DiD: estimated treatment effects for high-PMAY versus low-PMAY states by year relative to the 2015 PMAY launch (year 0, red dashed line). Pre-period coefficients (green background) are flat and statistically indistinguishable from zero ($F = 0.406$, $p = 0.805$), directly supporting the parallel trends assumption. Post-period coefficients (pink background) rise monotonically, reaching +1.5 log-mining units by year +9.

Instrumental Variables (2SLS). A key concern with the DiD design is reverse causality or selection: perhaps states that were already heavily mined received less PMAY allocation because of political dynamics, or mining-intensive states had pre-existing construction industries that attracted more PMAY investment. To address this, we estimate a Two-Stage Least Squares (2SLS) model using the central government’s financial assistance to each state (in crore rupees) as an instrument for PMAY construction scale.

An instrument is valid if it (1) strongly predicts the treatment variable (PMAY scale) and (2) affects the outcome (mining) only through its effect on the treatment, not through any independent channel. Central assistance is determined by a national-level formula based on urban population and housing deficit rather than by local mining conditions, making it plausibly exogenous to state-level mining propensity. The first stage confirms instrument strength ($F = 16.40$, $p < 0.001$). The 2SLS estimate of the PMAY effect on mining ($\hat{\beta}_{2SLS} = +0.631$) is slightly larger than the OLS estimate ($\hat{\beta}_{OLS} = +0.589$), with the Hausman difference (+0.042) suggesting that OLS may slightly underestimate the true causal effect, consistent with attenuation bias from measurement error in PMAY allocation figures.

Synthetic Control for Uttar Pradesh. As a complementary causal analysis, we construct a “synthetic Uttar Pradesh” (UP), a weighted combination of donor states that best matches UP’s pre-2015 mining trajectory, and compare actual post-2015 UP mining against this counterfactual. The pre-period fit is good (Root Mean Square Error = 0.12), with the

synthetic control drawing weights primarily from an unnamed central Indian state (72.4%), Arunachal Pradesh (11.7%), Madhya Pradesh (9.9%), and Maharashtra (6.0%). Post-2015, actual UP mining diverges from the synthetic control by an average of +0.948 log-mining units, translating to 158% excess mining relative to the counterfactual, consistent with the DiD estimate.

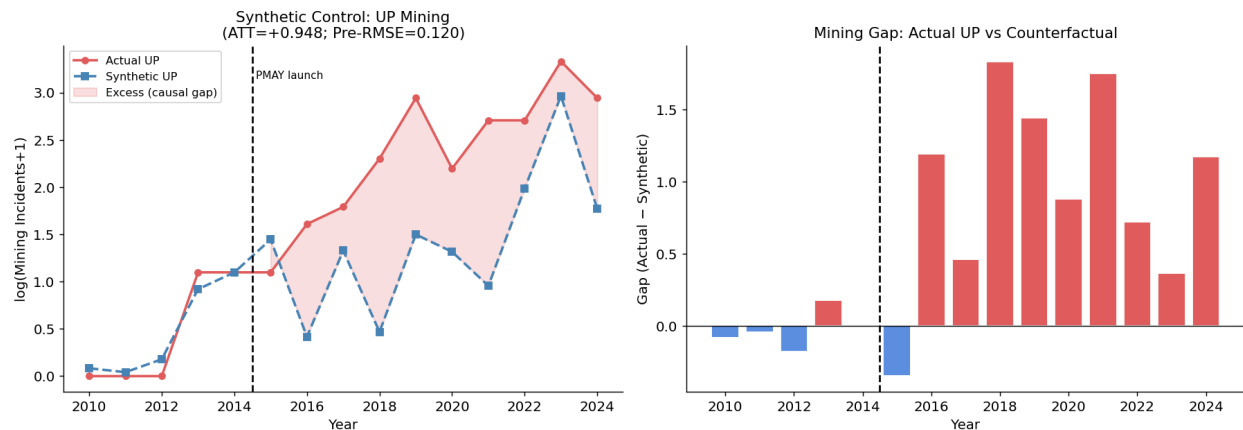
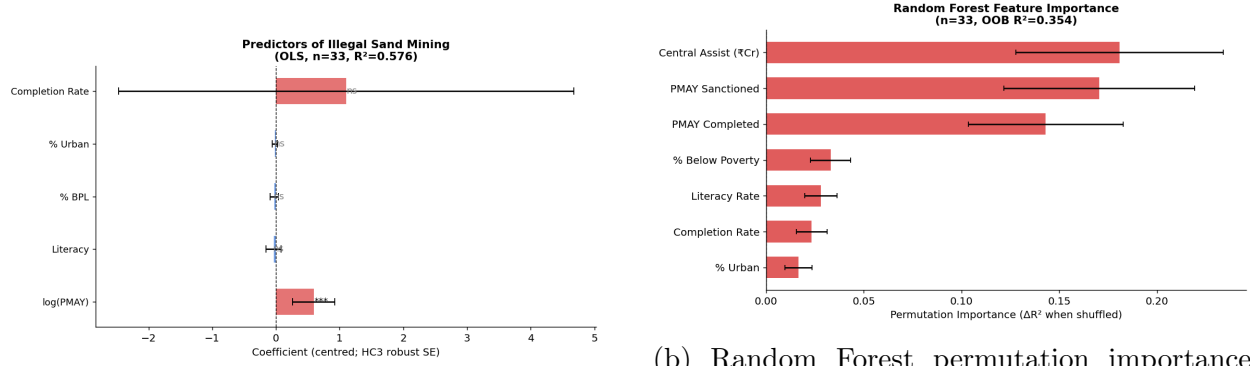


Figure 10: Synthetic control for Uttar Pradesh. Left: actual UP mining trajectory (red, solid) versus the synthetic counterfactual constructed from donor-state weights (blue, dashed); shaded region is the causal excess attributable to PMAY. Right: annual gap between actual and synthetic UP, consistently positive post-2015. Average treatment effect $ATT = +0.948$ log-mining units (+158% excess mining, Pre-RMSE = 0.12).

Mediation Analysis. Finally, we conduct a bootstrapped mediation analysis to quantify how much of poverty’s effect on mining is mediated by PMAY construction demand (the causal chain: poverty $\rightarrow$ PMAY allocation $\rightarrow$ mining). Using 5,000 bootstrap samples (random resamples of the data with replacement, used to estimate the uncertainty of indirect effects), we find that 63.6% of the total effect of poverty on mining is mediated through PMAY construction, though confidence intervals are wide and the individual paths are not individually significant at $p < 0.05$ in a dataset of 33 states. This is consistent with a demand-side interpretation: poorer states receive more PMAY funding, which increases construction activity, which increases demand for sand.

2.2.3 Socioeconomic Correlates (police__literacy__marginal_workers__agriculture.ipynb and household_and_crime.ipynb)

Police Strength and Mining Intensity. If illegal sand mining persists because of enforcement failure, we would expect states with lower police strength per capita to have higher mining intensity. The five states with the most mining news reports are Madhya Pradesh (279), Maharashtra (182), Tamil Nadu (120), Gujarat (114), and Andhra Pradesh (99). Computing a “mining incidents per police personnel” ratio reveals a striking pattern:



(a) OLS predictors of mining intensity ($n = 33$, $R^2 = 0.576$, HC3 robust SE). $\log(\text{PMAY})$ is the only statistically significant predictor (***) ; socioeconomic variables become non-significant once construction scale is controlled.

(b) Random Forest permutation importance for mining intensity ($n = 33$, OOB $R^2 = 0.354$). PMAY-related variables (central assistance, houses sanctioned, completed) dominate, substantially outweighing poverty rate, literacy, and urbanization.

Figure 11: Cross-sectional predictors of state-level illegal sand mining intensity from the construction demand analysis.

State	Mining Incidents	Mining per Police Officer
Madhya Pradesh	279	0.533
Gujarat	114	0.511
Maharashtra	182	0.034
Andhra Pradesh	99	0.019
Tamil Nadu	120	0.012

Madhya Pradesh and Gujarat show mining-to-police ratios roughly 15–45 times higher than Maharashtra, Tamil Nadu, and Andhra Pradesh. However, this comparison is complicated by the fact that high ratios could reflect either genuine enforcement failure or simply that those states have more mining activity to report in the first place. The authors note in the analysis that without district-level police deployment data (which is not publicly available), this finding is suggestive but not conclusive.

Housing Quality. A composite “housing quality index” combining the proportions of good and liveable housing in each state shows a strong negative correlation with mining intensity among the top five mining states: the correlation between $\log(\text{mining incidents})$ and the housing quality index is $r = -0.750$ (Pearson correlation, where -1 indicates perfect negative linear relationship and 0 indicates no relationship). States with poorer housing conditions tend to have more mining activity, consistent with the hypothesis that unmet housing demand is an underlying driver, both directly (through informal construction using illegally sourced sand) and indirectly (through PMAY allocation to poorer states).

Marginal Workers. The marginal worker ratio (the proportion of the labor force classified

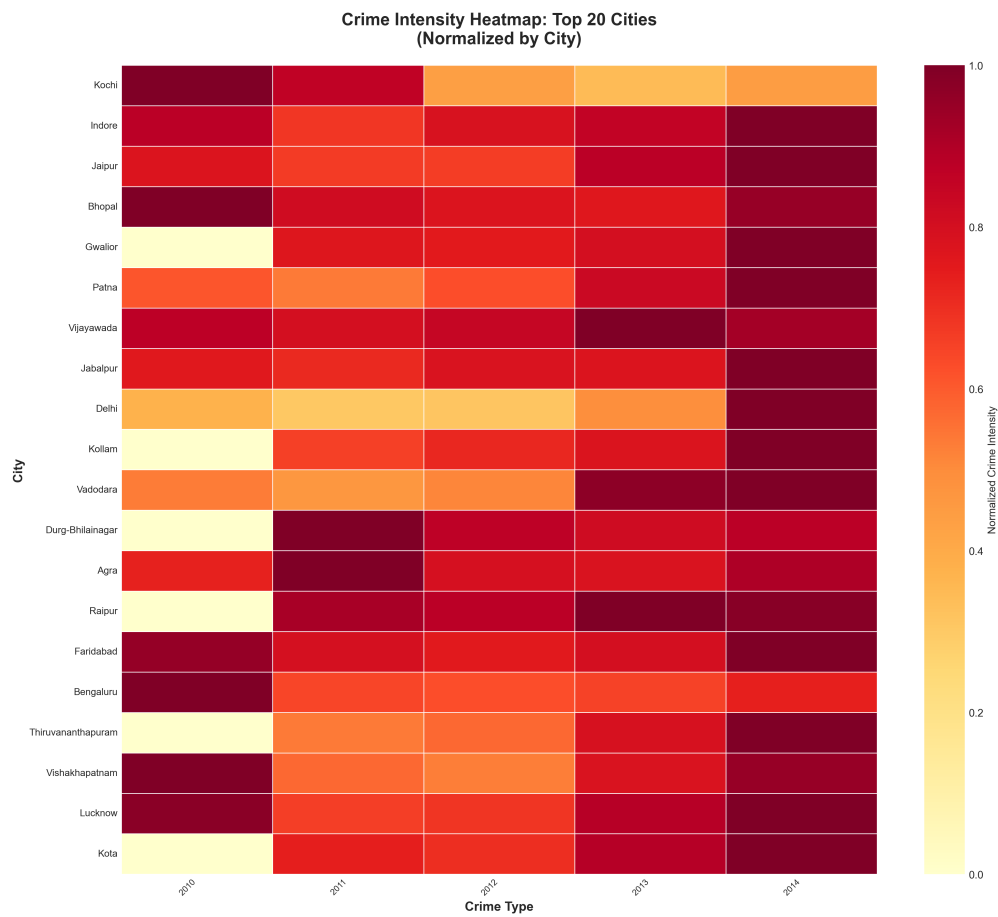
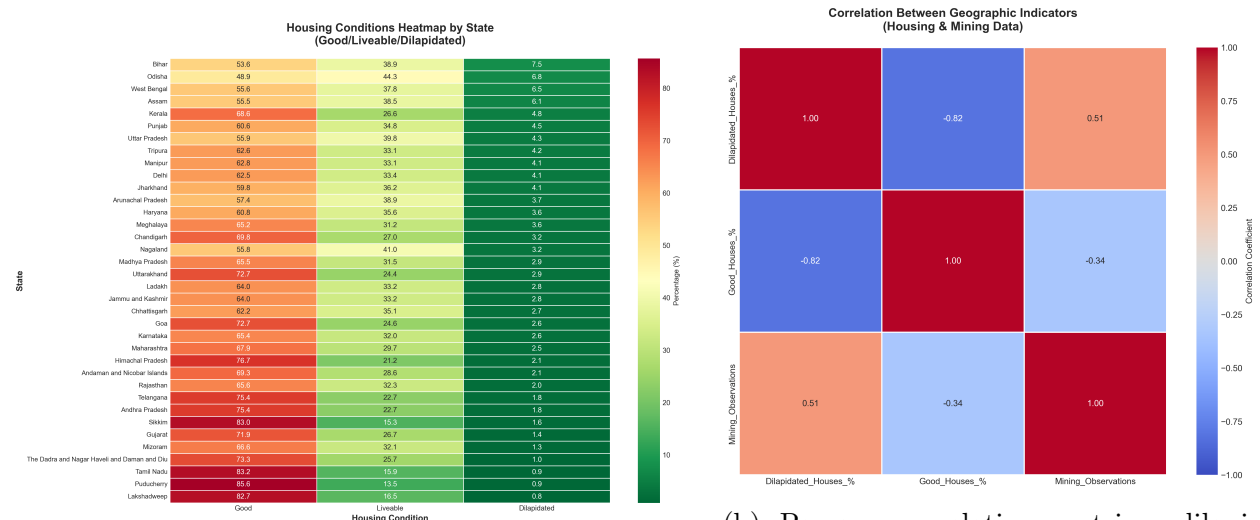


Figure 12: Normalized crime intensity heatmap for the top 20 Indian cities across 2010–2014. Cities in high-mining states (Gwalior, Patna, Jabalpur, Indore) show consistently high crime intensity. The multi-year pattern indicates structural rather than episodic enforcement environments.

as working fewer than 183 days per year) shows the strongest single-predictor correlation with mining intensity among socioeconomic variables: $r = +0.612$. States with higher proportions of marginally employed workers (those engaged in unstable, seasonal, or informal labor) show substantially more mining activity. This is consistent with the labor supply interpretation: illegal sand mining is physically demanding, relatively unskilled, and can be conducted seasonally (especially during dry months when river levels drop). It therefore represents an accessible income source for workers who lack stable employment alternatives. Literacy rate and agricultural dependence show weaker and statistically non-significant correlations with mining activity in this sample.

Housing Conditions and Crime. The correlation between state-level dilapidated housing rates and mining observations is $+0.507$ across all states, reinforcing the structural inequality dimension of the problem. The five states with the highest rates of dilapidated housing are Bihar (7.5%), Odisha (6.8%), West Bengal (6.5%), Assam (6.1%), and Kerala (4.8%): a list that overlaps substantially with high-mining states, though the overlap is not perfect (Kerala has high dilapidated housing but relatively low mining intensity, likely reflecting state-specific governance quality).



(a) Housing conditions by state (good, liveable, dilapidated). Bihar, Odisha, West Bengal, and Assam record the highest dilapidated proportions and are also among the top mining states.

(b) Pearson correlation matrix: dilapidated housing rate $r = +0.51$ with mining observations; good housing rate $r = -0.34$ with mining, forming a coherent structural inequality signal.

Figure 13: State-level housing conditions and their relationship with mining intensity.

2.2.4 NLP-Based Classification of Illegality (classification.ipynb)

A core analytical challenge is that not every observed mining incident in the India Sand Watch data is unambiguously illegal. Some observations may describe permitted operations

incorrectly identified as illegal; others may describe genuinely illegal mining without using legal terminology. We designed a rule-based scoring system to classify field observations by their probability of reflecting illegal activity, drawing on patterns derived from the court document corpus.

Feature Engineering. For each of the 375 field observations, we extract three classes of signals from the free-text “Notes from the location” field:

1. *Operational signals:* Binary indicators for whether the text mentions machinery (JCBs, excavators, pumps) and transport vehicles (trucks, tractors, trolleys). These are weighted most heavily (3 points each) because large-scale mechanized extraction is definitionally inconsistent with small-scale permitted mining under the Mines and Minerals (Development and Regulation) Act.
2. *Risk signals:* Binary indicators for nighttime activity, high-volume extraction, and repeated/daily operations (1 to 1.5 points each). These amplify the illegality score when present alongside operational signals.
3. *Legal signals:* Pattern matching against nine regular expression (text pattern) rules derived from legal language found in court documents: references to illegal mining, unauthorized operations, lack of environmental clearance, lease violations, and excess extraction (2 points each match).

Classification Thresholds. Observations scoring above the 80th percentile of the total score distribution are classified as *highly likely illegal*; those between the 50th and 80th percentiles are classified as *possibly illegal*. Of 375 observations, 121 (32%) are classified as highly likely illegal and 254 (68%) as possibly illegal.

Findings. The states with the highest absolute counts of highly likely illegal observations are Uttar Pradesh (32), Madhya Pradesh (30), West Bengal (25), and Bihar (19). When normalized by total observations per state, the highest proportions are in Goa (100%), Himachal Pradesh (100%), Rajasthan (87.5%), and Madhya Pradesh (69.8%). Bihar has the lowest proportion (9.8%) despite having the most total field observations, which is consistent with its large volume of lower-intensity, possibly artisanal (small-scale, manual) extraction that does not trigger the operational machinery signals.

Enforcement Gap Analysis. We compute, for each state, the gap between the rate of highly likely illegal observations and the rate of court-recognized illegal cases. States where observed illegal mining far exceeds court cases are those where enforcement is most severely lagging behind documented criminal activity. Himachal Pradesh, West Bengal, and Jharkhand show the largest positive gaps (more observed illegal activity than court activity), while Haryana and Karnataka show negative gaps (more court cases than field observations, possibly reflecting stronger proactive enforcement). This gap analysis directly identifies the

states where strengthened enforcement would have the highest expected impact.

2.2.5 Water Quality: Ganga and Sangam Sensor Analysis (spatial_analysis.ipynb, Section 11)

Two continuous monitoring sensors deployed at Prayagraj, Uttar Pradesh captured approximately 99,000 combined one-minute water quality readings between January 2019 and February 2020. We treat these sensors as a case study in what longitudinal, high-resolution environmental monitoring of mining impacts could look like at scale.

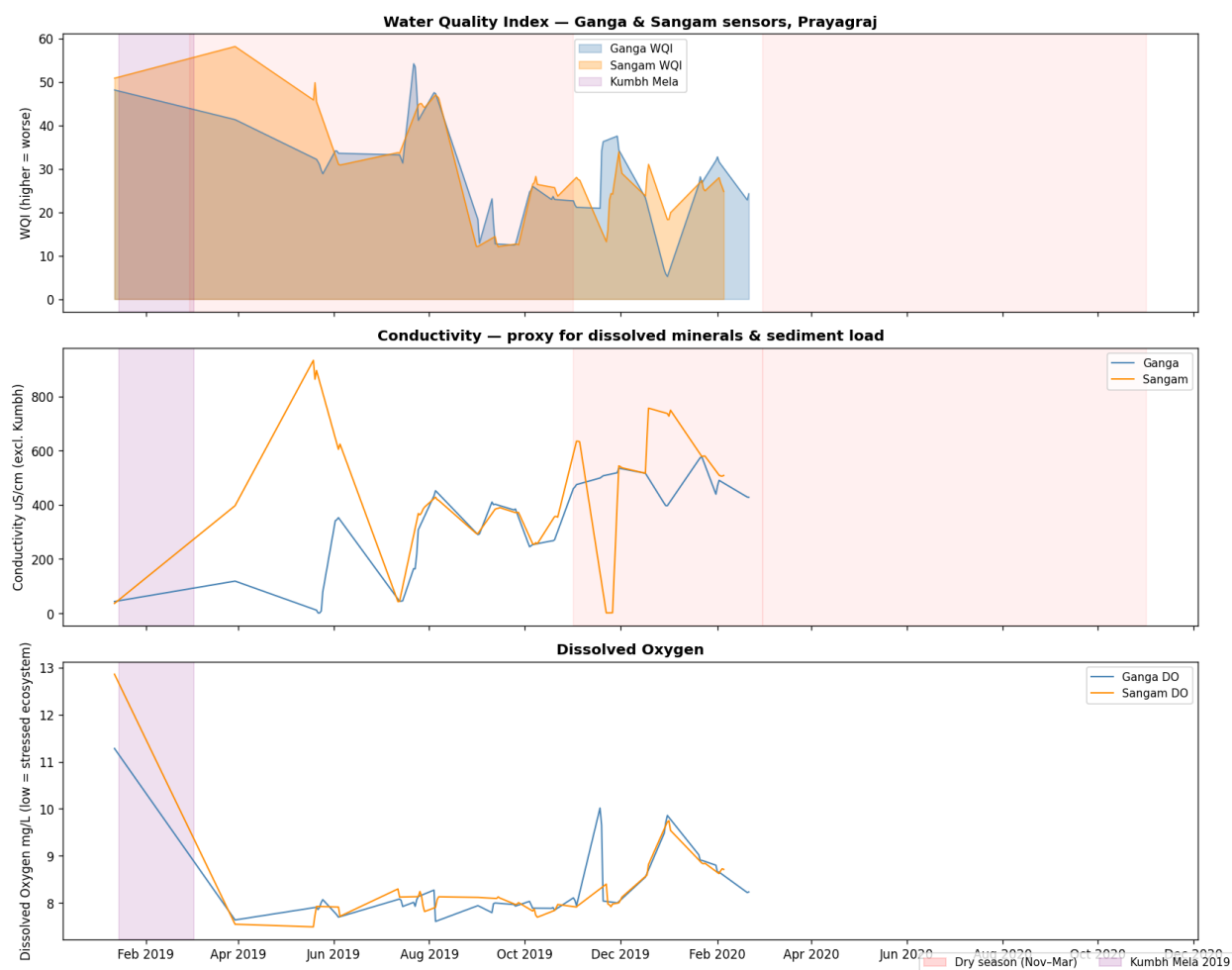


Figure 14: Full time-series overview for both sensors (January 2019 to February 2020). Top panel: Water Quality Index (WQI, higher = worse). Middle panel: conductivity ($\mu\text{S}/\text{cm}$), the primary proxy for riverbed mineral disturbance. Bottom panel: dissolved oxygen. The Kumbh Mela period (shaded pink, Feb–Mar 2019) and dry season (shaded beige) are marked. Both sensors track each other closely for most variables; the conductivity divergence reflects the Sangam sensor’s additional capture of the Yamuna river’s mineral load.

Seasonal Patterns. Our primary hypothesis is that river conductivity (a measure of dis-

solved mineral content and a proxy for riverbed disturbance) should peak during the dry season (November through February) when river levels fall, making riverbed sand accessible and allowing mineral-rich porewater to concentrate. Excluding the Kumbh Mela period (a major religious gathering that introduced confounding through exceptional crowd and sanitation effects), dry-season Ganga conductivity averages 478 $\mu\text{S}/\text{cm}$ versus 276 $\mu\text{S}/\text{cm}$ during the monsoon (June through September), a $1.73\times$ ratio with Cohen’s $d = 1.78$ (a very large effect by conventional standards) and a Mann-Whitney $p < 0.001$. A monthly Spearman correlation between conductivity and mining incidents shows $\rho = 0.536$ ($p = 0.089$), not quite reaching conventional significance in a dataset of only 12 monthly observations, but in the expected direction.

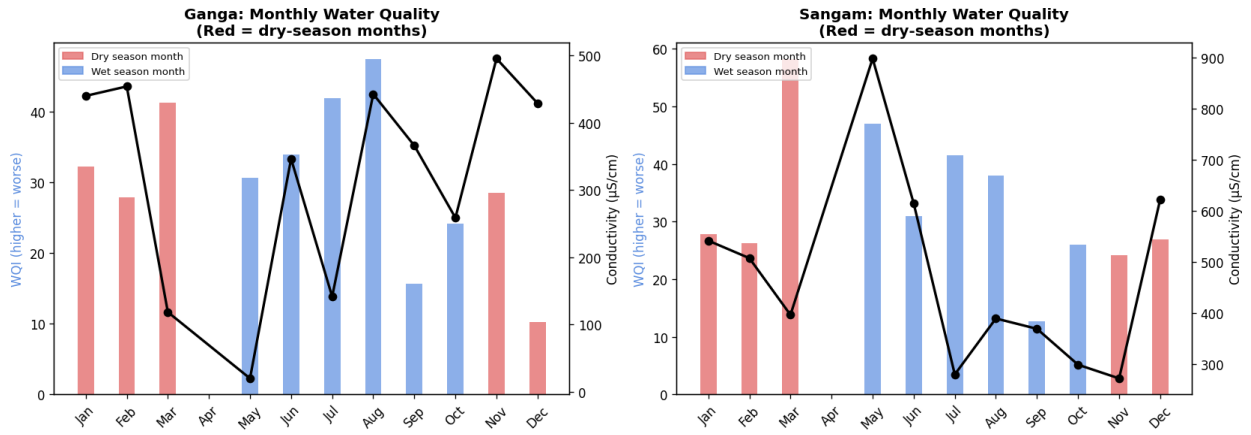


Figure 15: Monthly water quality at the Ganga (left) and Sangam (right) sensors. Red bars denote dry-season months (Nov–Feb); blue bars denote monsoon-season months. The black line (right axis) plots conductivity ($\mu\text{S}/\text{cm}$). Both sensors show higher WQI scores (worse quality index) and elevated conductivity during dry-season months, consistent with increased riverbed disturbance when river levels drop and sand becomes accessible.

Cross-Sensor Validity. The Ganga and Sangam sensors, located within a few kilometers of each other, show high cross-sensor correlations for most variables (WQI: $r = +0.891$, DO: $r = +0.973$, temperature: $r = +0.981$), validating that their readings reflect basin-wide water quality signals rather than sensor-specific artifacts. The exception is conductivity ($r = +0.372$), which shows higher divergence, likely because the Sangam sensor captures the combined mineral load of both the Ganga and Yamuna rivers, while the Ganga sensor captures a single river’s contribution.

Trend Decomposition. Seasonal decomposition of the Ganga conductivity time series (separating long-term trend, recurring seasonal patterns, and irregular residuals) reveals a positive overall trend of $+1.37 \mu\text{S}/\text{cm}$ per day across the monitoring period. The decomposed seasonal component shows a 30-day cyclical pattern consistent with tidal backflow effects. These methods are introduced here as proof-of-concept: a longer-term sensor deployment

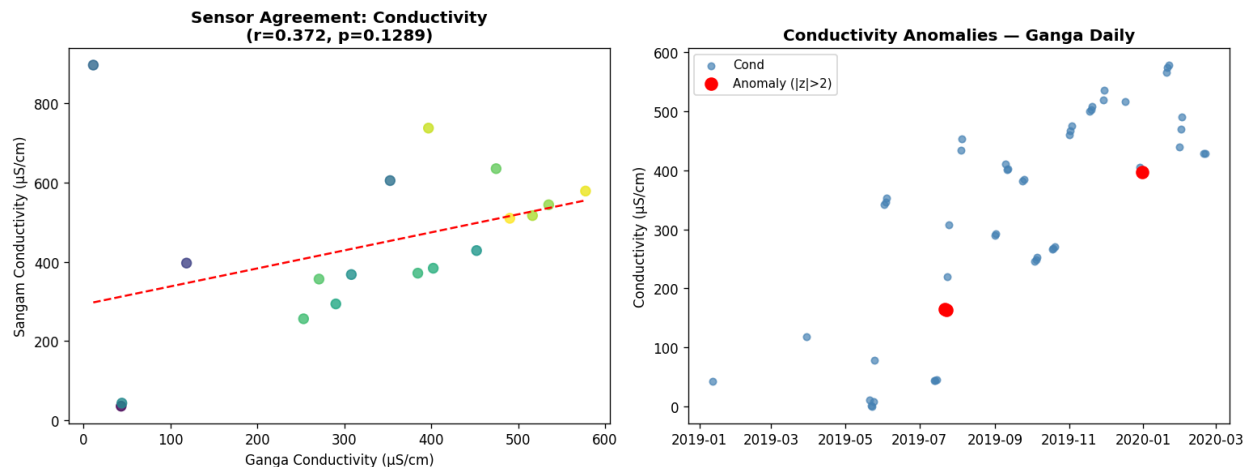


Figure 16: Left: cross-sensor conductivity scatter plot (monthly Ganga vs. Sangam averages, $r = +0.372$). The weaker cross-sensor agreement for conductivity compared to WQI ($r = +0.891$) and DO ($r = +0.973$) reflects the Sangam sensor’s additional capture of Yamuna mineral load. Right: daily Ganga conductivity time series with anomaly detection; red points ($|z| > 2$) flag statistical outliers in the series, both positive (high mineral days) and negative (sensor gaps or dilution events).

would allow more definitive attribution of conductivity trends to specific drivers including sand extraction.

2.3 Current State of the Issue

Across all five analytical modules, the following consolidated picture of illegal sand mining in India emerges.

Scale and distribution. Illegal sand mining is a national problem, documented across 32 states and Union Territories, but it is not uniformly distributed. Two geographic corridors account for the majority of confirmed incidents: the Gangetic plains (Bihar, eastern Uttar Pradesh, West Bengal) and the Central India plateau (Madhya Pradesh, Rajasthan, Goa, Gujarat). Within these corridors, mining is concentrated along specific rivers: the Ganga, Chambal, Narmada, Yamuna, and Mahananda are the most heavily extracted.

Growth trajectory. Documented mining incidents have grown by $14.7\times$ since 2015. While improved reporting infrastructure accounts for some of this increase, the causal evidence from the DiD and synthetic control analyses strongly implicates PMAY-U construction demand as a genuine driver of additional illegal extraction.

Enforcement failure. Crime rate (as a proxy for overall law enforcement quality) ranks as the third most important predictor of mining site location in the Random Forest model. The mining-to-police ratio analysis shows enormous variation across states. The enforcement

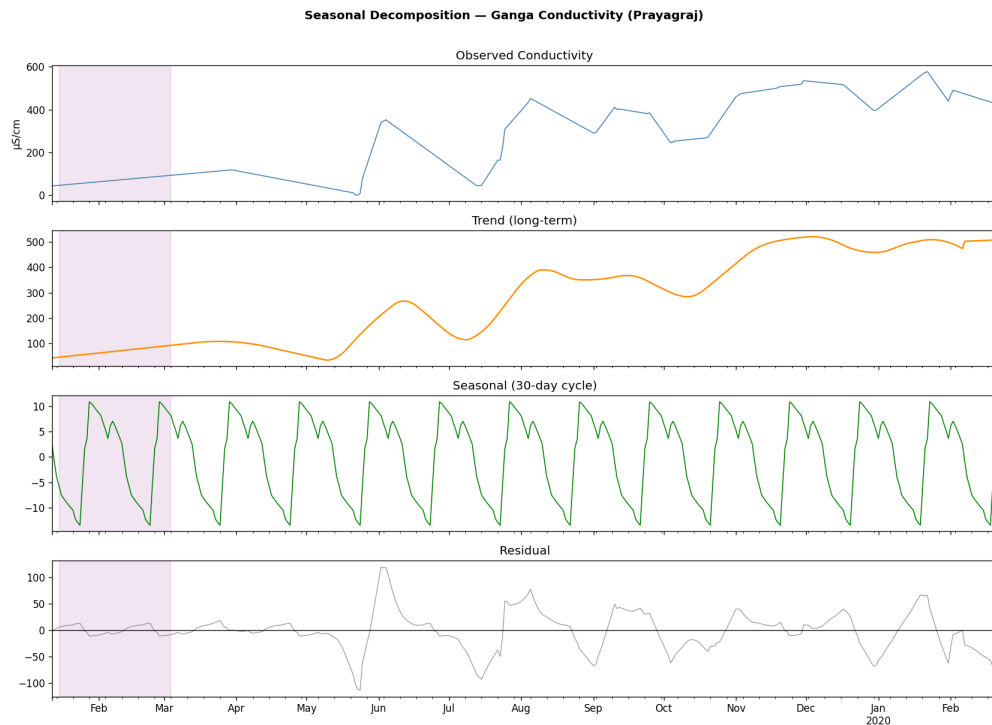


Figure 17: Seasonal decomposition of the daily Ganga conductivity time series (additive STL model). Top panel: observed values. Second panel: long-term trend ($+1.37 \mu\text{S}/\text{cm}/\text{day}$). Third panel: 30-day seasonal component consistent with tidal backflow modulation. Bottom panel: irregular residuals. The Kumbh Mela period (shaded, Feb–Mar 2019) is excluded from fitting due to exceptional confounding from the mass gathering.

gap analysis reveals that Himachal Pradesh, West Bengal, and Jharkhand have the largest imbalance between observed illegal activity and court recognition. These findings collectively indicate that enforcement failure is not incidental but structural.

Socioeconomic underpinning. Marginal employment rates, dilapidated housing proportions, and poverty rates all correlate positively with mining intensity. The mediation analysis suggests that poverty shapes mining activity through two channels: directly, by creating a labor supply of workers with few alternatives, and indirectly, by channeling government construction investment into states that then generate additional sand demand. This creates a cruel paradox: the government program intended to lift the poor out of inadequate housing is partly fueled by an industry that destroys the environmental resources on which poor communities depend.

Environmental damage. GWR analysis confirms that proximity to mining sites is associated with elevated groundwater TDS at 100% of Bihar monitoring stations and 78% of West Bengal stations. Seasonal conductivity analysis at Prayagraj confirms that dry-season water quality degradation aligns temporally and directionally with peak mining periods. PM10 is the strongest state-level predictor of mining intensity among environmental variables, consistent with the documented generation of coarse dust during sand transport.

3 Key Quantitative Findings and Significant Associations

This section consolidates the most statistically robust results across all analytical modules. We distinguish between causal findings (derived from identification strategies designed to isolate treatment effects) and associational findings (statistically significant correlations that may reflect confounding but are nonetheless informative for targeting).

3.1 Causal Evidence: Construction Demand Drives Illegal Mining

The Difference-in-Differences, Instrumental Variables, and Synthetic Control analyses converge on a consistent and robust conclusion: the 2015 launch of PMAY-U caused a measurable and substantial increase in illegal sand mining in high-PMAY states. Four independent identification strategies reach the same directional finding.

- **Difference-in-Differences:** $\hat{\beta} = +1.057$ ($p < 0.001$, 95% CI: [0.855, 1.259]). High-PMAY states experienced **187.7% more mining** post-2015 relative to low-PMAY states, after removing all state-level and year-level confounds through fixed effects. This is a large, precisely estimated treatment effect.

- **Event Study pre-trend test:** $F = 0.406$, $p = 0.805$. No significant pre-2015 differential trends between treated and control states, directly supporting the parallel trends assumption on which DiD validity rests.
- **Two-Stage Least Squares (Instrumental Variables):** $\hat{\beta}_{2SLS} = +0.631$, first-stage $F = 16.40$ ($p < 0.001$). Using central government financial assistance as an instrument for PMAY scale (an allocation determined by national population and housing-deficit formulas rather than local mining conditions), the 2SLS estimate exceeds the OLS estimate (+0.589), consistent with attenuation bias from measurement error.
- **Synthetic Control for Uttar Pradesh:** Actual post-2015 UP mining exceeded the synthetic counterfactual by +0.948 log-mining units, translating to **158% excess mining** relative to what UP would have experienced absent PMAY. Pre-treatment Root Mean Square Error = 0.12, confirming a good counterfactual fit.
- **Bootstrapped Mediation:** **63.6%** of poverty’s total effect on mining is mediated through PMAY construction demand (5,000 bootstrap iterations). This quantifies the full causal chain: poverty $\rightarrow$ PMAY allocation $\rightarrow$ sand demand $\rightarrow$ mining.

3.2 Spatial Finding: Groundwater Contamination Near Mining Sites

Geographically Weighted Regression applied to 7,190 WRIS monitoring stations reveals a highly spatially heterogeneous contamination signal. The global OLS model explained only 15.8% of variance; allowing coefficients to vary spatially raises explained variance to 54.5%, confirming that the relationship between mining proximity and water quality is fundamentally local.

- **Bihar: 100% of monitoring stations** show negative local GWR coefficients (closer to active mining is associated with higher Total Dissolved Solids), with median $\beta = -0.346$. Bihar’s shallow alluvial Gangetic geology connects the water table directly to the mined riverbed, producing a strong and consistent contamination signal.
- **West Bengal: 78% of monitoring stations** show negative coefficients, with a stronger median effect ($\beta = -0.505$) than Bihar. The Mahananda and lower Ganga tributaries carry heavy extraction activity; the higher per-unit magnitude suggests West Bengal’s monitoring stations are more consistently co-located with intensive mining zones.
- **Madhya Pradesh:** Only 39% negative coefficients (median $\beta = +0.215$). The Chambal basin’s hard-rock geology produces naturally high mineral groundwater regardless of mining activity, masking the anthropogenic signal.

- **Global GWR** $R^2 = 0.545$, versus OLS $R^2 = 0.158$: a $3.4\times$ improvement from allowing spatial heterogeneity.

3.3 Socioeconomic Associations

- **Marginal worker ratio:** $r = +0.612$ with mining intensity (Pearson). The single strongest socioeconomic predictor. States with larger proportions of informally employed workers (fewer than 183 days per year) have substantially more mining activity, consistent with a labor-supply interpretation: illegal sand mining offers accessible seasonal income to workers who lack stable alternatives.
- **Housing quality index:** $r = -0.750$ among the top five mining states. States with poorer aggregate housing conditions have more mining, consistent with both unmet local construction demand and the geographic concentration of vulnerable communities near mining rivers.
- **Dilapidated housing rate:** $r = +0.507$ across all states. Bihar (7.5% dilapidated), Odisha (6.8%), West Bengal (6.5%), and Assam (6.1%) are both high-mining and high-dilapidated-housing states, reinforcing the structural inequality dimension of the problem.
- **Police-to-mining ratio variation:** Madhya Pradesh and Gujarat show mining-per-officer ratios **15 to 45 times higher** than Maharashtra, Tamil Nadu, and Andhra Pradesh, suggesting large disparities in the enforcement environment across states.

3.4 Environmental Time-Series Findings

- **Dry-season conductivity elevation:** Dry-season (November through February) average conductivity at the Ganga sensor is **478 $\mu\text{S}/\text{cm}$** versus 276 $\mu\text{S}/\text{cm}$ during monsoon (June through September): a **1.73 $\times$ ratio** with Cohen's $d = 1.78$ (large effect by conventional standards) and Mann-Whitney $p < 0.001$. This aligns precisely with peak mining periods when low river levels expose riverbed sand.
- **Cross-sensor agreement:** WQI cross-sensor correlation $r = +0.891$, DO $r = +0.973$, temperature $r = +0.981$. Both buoys reflect the same basin-wide water quality signal, validating the dataset's reliability. Conductivity diverges ($r = +0.372$) because the Sangam sensor captures the combined Ganga-Yamuna mineral load.
- **Conductivity trend:** $+1.37 \mu\text{S}/\text{cm}$ per day across the monitoring period, consistent with progressive riverbed disturbance over the 13 months.
- **Monthly conductivity-mining correlation:** Spearman $\rho = 0.536$, $p = 0.089$. Directionally consistent with the hypothesis that mining peaks drive conductivity peaks;

not statistically significant at $p < 0.05$ in a dataset of only 12 monthly observations.

3.5 Classification and Enforcement Gap

- Of 375 field observations, **121 (32%) are classified as highly likely illegal** and 254 (68%) as possibly illegal, using a weighted scoring system based on machinery use, nighttime activity, volume indicators, and legal language patterns extracted from 91 court documents.
- Highest absolute counts of highly likely illegal observations: Uttar Pradesh (32), Madhya Pradesh (30), West Bengal (25), Bihar (19). Highest proportional rates: Goa (100%), Himachal Pradesh (100%), Rajasthan (87.5%), Madhya Pradesh (69.8%).
- Largest **enforcement gaps** (observed illegal activity exceeding court-recognized cases): Himachal Pradesh, West Bengal, and Jharkhand. These are the states where strengthened enforcement would have the highest expected marginal impact.
- **Random Forest** site-location classifier: ROC-AUC = 0.965 ± 0.014 (5-fold cross-validation). Feature importance ranking places longitude first, latitude second, crime rate third, PM10 fourth, PM2.5 fifth. Crime rate outranks both air quality measures, indicating that the enforcement environment is at least as strong a determinant of mining location as environmental conditions.

4 Recommendations and Policy Suggestions

Based on the empirical evidence developed in this analysis, we offer the following policy recommendations, organized by institutional actor.

4.1 For the Central Government

Decouple PMAY-U sand demand from illegal sources. The causal evidence from our DiD, IV, and synthetic control analyses converges on a finding that large-scale centrally-sponsored housing construction generates demand that is being partially met by illegal extraction. The central government should require state governments to source sand for PMAY projects exclusively from verified legal stockpiles, and should establish a Sand Sourcing Compliance Audit as a condition of central assistance release. This transforms a demand-side driver of illegal mining into a demand-side enforcement lever.

Establish a national sand price floor and distribution system. A significant driver of illegal sand's attractiveness is its price: illegal sand sold without royalties and transportation duties is substantially cheaper than legally extracted sand. A centrally-set price floor for

legally mined sand, combined with state-managed distribution depots (similar to the cement price control boards that existed historically), would narrow the price differential that makes illegal supply competitive.

Fund continuous environmental monitoring in high-mining corridors. The Ganga/Sangam sensor analysis demonstrates the feasibility and value of continuous, automated water quality monitoring. We recommend deploying continuous conductivity and TDS monitoring buoys at 20 to 30 sites along the Ganga, Chambal, and Narmada rivers, with data publicly accessible in real time. This would create an objective, legally admissible evidentiary record of environmental damage linked to specific time periods and locations.

4.2 For State Governments

District-level police deployment in high-gap states. Our enforcement gap analysis identifies Himachal Pradesh, West Bengal, Jharkhand, Gujarat, and Goa as states where observed illegal mining greatly exceeds court-recognized cases. State governments in these jurisdictions should deploy dedicated sand mining enforcement cells at the district level, with specific performance metrics (cases filed per observed incident) tied to district collector accountability.

Implement machine-verified transport permits. A large portion of illegal sand escapes detection not at the extraction point but during transport. A GPS-tracked, QR-code-verified e-permit system for sand transport vehicles (modeled on the e-way bill system used for goods transport under GST) would allow enforcement agencies to intercept illegal shipments at checkpoints without needing to monitor every river stretch. Several states, including Rajasthan and Madhya Pradesh, have piloted versions of such systems; these should be made mandatory and standardized.

Alternative livelihood programs for marginal workers. The correlation between marginal worker ratios and mining intensity points to a labor supply dimension of the problem. Sand mining offers seasonal income to workers who lack stable alternatives. State employment guarantee programs (under the Mahatma Gandhi National Rural Employment Guarantee Act, MGNREGA) should be expanded and prioritized in high-mining districts during the dry-season months (October through February) when mining intensity peaks. This directly competes with illegal mining for the available labor pool.

4.3 For Judicial and Regulatory Bodies

Standardize NGT enforcement orders. Our court document analysis finds that NGT orders are the most common form of legal response to illegal sand mining, but their enforcement is highly variable. The NGT should issue standardized compliance affidavit re-

quirements and appoint state-level monitoring committees with fixed reporting timelines, reducing the discretion of individual district administrators in implementing environmental orders.

Treat sand mining mafia cases under organized crime legislation. Currently, most sand mining prosecutions proceed under the Mines and Minerals (Development and Regulation) Act or environmental protection statutes, which carry relatively light penalties. The organized, cross-district, and often interstate nature of Sand Mafia operations (as evidenced by the HH spatial clustering of Goa, Gujarat, Madhya Pradesh, and Rajasthan) qualifies these networks for prosecution under the Maharashtra Control of Organised Crime Act (MCOCA) and its state-level equivalents. This would enable asset forfeiture, longer custody periods, and stronger deterrence.

4.4 For Civil Society and Research

Expand the India Sand Watch observer network. Our data quality analysis reveals severe geographic gaps in field observation coverage: northeastern states, Odisha, Chhattisgarh, and Kerala are almost entirely absent from the mining observation dataset despite having significant rivers. India Sand Watch should invest in trained observer networks in these states, particularly targeting riverine communities directly affected by extraction.

Longitudinal panel datasets for causal inference. The primary limitation of our causal analyses is the cross-sectional nature of most available data. A district-level, annually updated panel dataset combining mining incidents, police deployment, PMAY completion rates, and environmental quality would allow much stronger causal identification using panel fixed effects methods. We encourage NDAP and state data agencies to prioritize the assembly of such a panel.

Limitations

We acknowledge several important limitations of this analysis. First, the mining observation data suffer from reporting bias: regions with more active civil society organizations, better internet connectivity, and more sympathetic local governance are more likely to have observations recorded. This means our spatial findings probably undercount mining in more remote or politically sensitive areas. Second, the causal analyses rely on state-level aggregations with small effective sample sizes (33 states for the cross-sectional OLS, and a limited number of pre-treatment periods for the DiD event study). While we have implemented robustness checks (sensitivity analysis excluding the three most data-heavy states, 2SLS to address endogeneity), these limitations bound the strength of causal claims we can make. Third, the GWR and water sensor analyses are observational: we cannot definitively attribute elevated

TDS or seasonal conductivity changes to sand mining specifically, as opposed to other seasonal and geographic factors. Fourth, no temporal causality can be claimed from purely cross-sectional correlations between socioeconomic variables and mining intensity.

References

- [1] United Nations Environment Programme (UNEP). *Sand and Sustainability: Finding New Solutions for Environmental Governance of Global Sand Resources*. UNEP, Geneva, 2019.
- [2] United Nations. *World Urbanization Prospects: The 2018 Revision*. Department of Economic and Social Affairs, Population Division, New York, 2018.
- [3] Veditum India Foundation. *India Sand Watch: Court Documents Database*. Available at: <https://indiawater.org/sandwatch>. Accessed 2024.
- [4] S. Dharmadhikary and M. Thakkar. *Sand Mining: The Unexamined Threat to Indian Rivers*. Manthan Adhyayan Kendra, Badwani, 2018.
- [5] National Crime Records Bureau (NCRB). *Crime in India 2022*. Ministry of Home Affairs, Government of India, New Delhi, 2023.
- [6] Ministry of Housing and Urban Affairs. *Pradhan Mantri Awas Yojana-Urban: Progress Report*. Government of India, New Delhi, 2024.
- [7] Central Pollution Control Board (CPCB). *National Ambient Air Quality Status and Trends*. Ministry of Environment, Forest and Climate Change, Government of India, New Delhi, 2023.
- [8] Ministry of Jal Shakti. *Water Resources Information System: Ground Water Quality Data*. Government of India, New Delhi, 2023.
- [9] Office of the Registrar General and Census Commissioner, India. *Census of India 2011: Primary Census Abstract*. Ministry of Home Affairs, Government of India, New Delhi, 2011.